Title: Identifying key healthcare assistance needs for older adults in Maryland.
Prepared for: Maryland State Government Officials

Summary:

As Maryland's older adult population grows, so does the urgency of addressing their complex healthcare needs. We analyzed data of over 53,000 patients to identify where the state can have the greatest impact within budget constraints. Focusing on patients aged 65 and older—who make up 73.6% of the dataset, we examined healthcare expenses, coverage, medical conditions, treatments, and geographic disparities. Based on our analysis, here are the key areas where the state of Maryland could provide healthcare assistance to the elderly population.

- Top conditions affecting elderly: This analysis identified the most common health conditions affecting Maryland's elderly population. These are not only common but often lead to expensive procedures later. By supporting preventive care, Maryland could reduce long-term costs and improve quality of life.
- Medication Assistance Programs: We identified commonly prescribed medications with high out-of-pocket costs. The state could implement medication assistance programs targeting these specific medications, particularly for patients with limited coverage.
- Geographic Focus: The county-level analysis identified ten counties with the highest average gaps between healthcare expenses and insurance coverage. Targeted county-level interventions such as mobile clinics, subsidies, or community care support could deliver outsized benefits.
- Targeted Demographic Assistance: Using machine learning, we identified four distinct groups among elderly patients. One group represents individuals over 90 years old with very high healthcare expenses and extremely low insurance coverage. The state could design targeted assistance programs for each cluster, focusing on their specific healthcare challenges.
- Age-Based Support Programs: This analysis showed that healthcare coverage ratios vary significantly by age group. The state could design graduated

assistance programs that provide more support to age groups with the largest coverage gaps.

- Partner with providers and insurers: To improve coverage for high-burden medications and essential procedures.

Key Findings:

- The average healthcare expense for elderly patients exceeds $1.27 million, yet insurance covers only 1–4% of that cost on average.
- Most common conditions among older adults:
    - Viral sinusitis (disorder) (count-43745)
    - Acute viral pharyngitis (disorder) (count-24046)
    - Body mass index 30+ - obesity (finding) (count-23089)
    - Prediabetes (count-21167)
- Top medications prescribed to elderly drive significant out-of-pocket expenses:
    - Hydrochlorothiazide 25 MG Oral Tablet
    - lisinopril 10 MG Oral Tablet
    - amLODIPine 2.5 MG Oral Tablet
- Most Expensive Procedures for Elderly Patients with average cost:
    - Lung Transplant ($ 514984.44)
    - Admit to ICU (procedure) ($ 264601.84)
    - Transplant of lung (procedure) ($ 175841.81)
- The age group analysis shows that the age group having the largest coverage gap is 90-94 years.
- Clustering analysis revealed specific subgroups of elderly patients with extremely low coverage, often despite very high needs.

- Geographic Disparities in Healthcare Utilization: The analysis of patient county reveals disproportionately large healthcare coverage gaps, with Charles, Queen Anne's, and Howard Counties being the most affected.

Proposed Intervention Areas:

Priority Area	Target Population	Potential Impact
Chronic condition support	Older adults with two or more conditions	Fewer hospitalizations, lower costs
Geographic focus zones	Select high-need counties	Equity in care, better outcomes
Preventive care access	All adults above 65	Healthier aging, early detection

Conclusion:

This analysis has identified several key opportunities for the state of Maryland to provide targeted healthcare assistance to its elderly population. By focusing on specific conditions, medications, geographic areas, and demographic groups with the largest coverage gaps, the state can stretch its limited resources further and deliver help where it’s needed most.